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HEAD-UP DISPLAY DEVICE AND CONTROL METHOD THEREOF

Abstract

A head-up display device includes two systems each having a backlight including light sources, a display that causes illumination light of the light sources to pass therethrough to generate display light, and a control unit that controls turning on and off of the light sources, in which, when a virtual image is displayed, a second control unit performs control such that at least some of second light sources are turned on, and, when a real image is displayed, a first control unit performs control such that at least some of first light sources are turned on, and performs control such that the power consumption of the first light sources per unit area of a first display is made to be larger than the power consumption of the second light sources per unit area of a second display when the virtual image is displayed.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a head-up display device and the like that have an emission port and emit display light from the emission port toward a translucent member, thereby causing a real image and a virtual image as display images represented by the display light to be visually recognized.

BACKGROUND ART

[0002] For example, FIG. 1 of Patent Document 1 describes a head-up display device (hereinafter, referred to as a HUD device **1**) that has an emission port **17** and emits display light from the emission port **17** toward a translucent member WS, thereby causing at least a real image RI as a display image represented by the display light to be visually recognized. According to the HUD device **1**, the HUD device **1** includes a first display element, and has a first display unit **12a** that causes light emitted from a first light source **11a** for the real image RI to pass therethrough and displays the real image RI of the display light, and a reflection unit **13** that reflects at least a first light beam L**1** representing the real image RI displayed on the first display unit **12a** toward the translucent member WS, in which the reflection unit **13** includes a first mirror **131** that reflects the first light beam L**1** toward a second mirror **132** and a third mirror **133** that reflects the first light beam L**1** toward the emission port **17**, the first mirror **131** is arranged, along an optical path of the first light beam L**1**, closer to the emission port **17** than the first display unit **12a** and closer to the first display unit **12a** than a first optical focal point F**1** of an imaging optical system including the translucent member WS, the second mirror **132**, and the third mirror **133**, and the third mirror **133** and the first display unit **12a** are arranged such that at least a position of the third mirror **133** in the height direction is above a position of the first display unit **12a** in the height direction. The HUD device **1** is described in Patent Document 1 (refer to the paragraph [0007]) unpublished as of the filing date of the present application.

[0003] In addition, FIG. 5 of Patent Document 1 illustrates a configuration in which display is performed by switching between a virtual image VI and the real image RI in the above-described HUD device **1**. According to FIG. 5, in addition to the configuration illustrated in FIG. 1, the HUD device **1** includes a second display element, and further has a second display unit **12b** that causes light emitted from a second light source **11b** for the virtual image VI to pass therethrough and displays the virtual image VI as a display image, in which the second mirror **132** configured by, for example, a half mirror causes a second light beam L**2** representing the virtual image VI displayed on the second display unit **12b** to pass therethrough, the third mirror **133** reflects the second light beam L**2** toward the windshield WS, and the second display unit **12b** is arranged, along an optical path of the second light beam L**2**, closer to the emission port **17** than a position of a second optical focal point F**2** of the imaging optical system including the windshield WS, the second mirror **132**, and the third mirror **133**, thereby causing the virtual image VI as a display image to be capable of being visually recognized by an occupant DR. Therefore, the HUD device **1** that performs display by switching between the virtual image VI and the real image RI can be realized (refer to the paragraphs [0038] and [0039]).

PRIOR ART DOCUMENT

Patent Document

[0004] Patent Document 1: Japanese Patent Application No. 2023-213466

SUMMARY OF INVENTION

Technical Problem

[0005] According to the technique described in Patent Document 1, the display image is displayed while switching between the real image RI and the virtual image VI by changing the optical paths (the first light beam L1 and the second light beam L2) of the imaging optical system including the first mirror **131**, the second mirror **132** (half mirror), and the third mirror **133**. An illumination optical system such as a lens for displaying a real image and an illumination optical system such as a lens for displaying a virtual image are exclusively designed, and the light source of a backlight for displaying a real image emits illumination light such that the illumination light converges to the first display unit **12a**, and the light source of a backlight for displaying a virtual image emits illumination light such that the illumination light diverges to the second display unit **12b**. Thus, it is known that the display light is darker when the real image is displayed than when the virtual image is displayed. Therefore, when the display image is displayed after switching from the real image to the virtual image or from the virtual image to the real image, the brightness of the display image is changed, and the occupant who is a viewer may feel discomfort.

[0006] Therefore, it is an object of the present invention to provide a head-up display device and the like capable of reducing discomfort of a viewer due to a change in brightness generated when a display image is displayed after switching between a real image and a virtual image.

[0007] Other objects of the present invention will become apparent to those skilled in the art by referring to the aspects and the best mode exemplified below, and the accompanying drawings.

Solution to Problem

[0008] Hereinafter, in order to easily understand the outline of the present invention, aspects according to the present invention will be exemplified.

[0009] A first aspect according to the present invention is a head-up display device that has an emission port and emits display light from the emission port toward a translucent member, thereby causing a real image and a virtual image as display images represented by the display light to be visually recognized, the head-up display device including: a first backlight including a first light source; a first display that causes illumination light of the first light source to pass therethrough to generate first display light; a second backlight including a second light source; a second display that causes illumination light of the second light source to pass therethrough to generate second display light; an imaging optical system including a first reflection member that reflects the first display light incident from one surface and causes the second display light incident from the other surface to pass therethrough, and a second reflection member that reflects the first display light reflected by the first reflection member or the second display light that has passed through the first reflection member to guide the first display light or the second display light to the emission port, in which the first display light is emitted along a first optical path that is reflected by the first reflection member to generate the real image, and the second display light is emitted along a second optical path that passes through the first reflection member and is reflected by the second reflection member to form the virtual image; a first control unit that controls turning on and off of a plurality of the first light sources of the first backlight; and a second control unit that controls turning on and off of a plurality of the second light sources of the second backlight, in which, when the virtual image is displayed, the second control unit performs control such that at least some of the second light sources are turned on, and, when the real image is displayed, the first control unit performs control such that at least some of the first light sources are turned on, and performs control such that power consumption of the first light sources per unit area of the first display is made to be larger than power consumption of the second light sources per unit area of the second display when the virtual image is displayed.

[0010] The first aspect is configured to include the first display that causes illumination light emitted from the first light source of the first backlight for displaying the real image to be reflected by the first reflection member and emits the illumination light toward the translucent member via the second reflection member (first optical path) and generates the real image as the first display

light, and the second display that causes illumination light emitted from the second light source of the second backlight for displaying the virtual image to pass through the first reflection member and emits the illumination light toward the translucent member via the second reflection member (second optical path) and generates the virtual image as the second display light, in which, when the virtual image is displayed, the second control unit performs control such that at least some of the second light sources are turned on, and, when the real image is displayed, the first control unit performs control such that at least some of the first light sources are turned on, and performs control such that the power consumption of the first light sources per unit area of the first display is made to be larger than the power consumption of the second light sources per unit area of the second display when the virtual image is displayed.

[0011] Here, “when the real image is displayed, performing control such that the power consumption of the first light sources per unit area of the first display is made to be larger than the power consumption of the second light sources per unit area of the second display when the virtual image is displayed” means, for example, (1) when the real image is displayed, performing control such that the value of the current (current value) flowing through the first light source of the first backlight is increased as compared to when the virtual image is displayed, (2) when the real image is displayed, performing control such that the number of the first light sources to be turned on is increased as compared to when the virtual image is displayed, or (3) when the real image is displayed, performing control such that the current value of the current flowing through the first light source is increased, and performing control such that the number of the first light sources to be turned on is increased as compared to when the virtual image is displayed, or when the real image is displayed, performing control such that the mounting density of the first light sources to be turned on on the outer edge side in the long-side direction of the light-source circuit mounting substrate formed of a rectangular region on which the first light sources are mounted is made to be larger than the mounting density of the second light sources to be turned on in an inner region of the outer edge side as compared to when the virtual image is displayed.

[0012] Thus, according to the first aspect, when the virtual image is displayed, the second control unit turns on all the second light sources of the second backlight, and, when the real image is displayed, the first control unit turns on all the first light sources of the first backlight and makes the power consumption of the first light sources per unit area larger than that when the virtual image is displayed by increasing the value of the current (current value) flowing through one first light source, so that the brightness when the real image is displayed can be improved, as a result, the difference (change) in the brightness when the display is switched to the virtual image or the real image can be reduced, and the discomfort of the occupant who is a viewer when the display is switched can be suppressed. Moreover, the second control unit may turn on some of the second light sources of the second backlight when the virtual image is displayed, and the first control unit may turn on all the first light sources of the first backlight when the real image is displayed. In this case, the brightness of the display images of the light sources may be the same in the real image display and the virtual image display, or control such that the light sources are turned on more brightly when the real image is displayed may be performed. Furthermore, according to the first aspect, since the light sources do not need to be shared between the virtual image and the real image, lenses in consideration of the light distribution characteristics of both the virtual image and the real image can be optimally designed. Accordingly, appropriate brightness and uniformity can be obtained in both the virtual image display and the real image display, and the display quality can be improved.

[0013] In a second aspect depending from the first aspect, when the real image is displayed, the first control unit may perform control such that a current value of a current flowing through the first light source is increased as compared to when the virtual image is displayed by the second control unit.

[0014] In the second aspect, when the real image is displayed, the first control unit performs

control such that the value of the current (current value) flowing through the first light source of the first backlight is increased as compared to when the virtual image is displayed by the second control unit. Thus, particularly, when the display is switched from the virtual image to the real image, the difference (change) in the brightness from when the virtual image is displayed can be reduced without changing the lighting area of the first light sources mounted on the first light-source circuit substrate of the first backlight, and the discomfort of the occupant who is a viewer when the display is switched can be suppressed.

[0015] In a third aspect depending from the first aspect, when the real image is displayed, the first control unit may perform control such that the number of the first light sources to be turned on is increased as compared to when the virtual image is displayed by the second control unit.

[0016] In the third aspect, when the display is switched from the virtual image to the real image, the first control unit increases the number of the first light sources to be turned on, which are mounted on the first light-source circuit substrate of the first backlight. Thus, the difference (change) in the brightness when the real image is displayed and when the virtual image is displayed can be reduced, and the discomfort of the occupant who is a viewer when the display is switched can be suppressed.

[0017] In a fourth aspect depending from the first aspect, when the real image is displayed, the first control unit may perform control such that a current value of a current flowing through the first light source is increased, and may perform control such that the number of the first light sources to be turned on is increased as compared to when the virtual image is displayed by the second control unit.

[0018] In the fourth aspect, when the display is switched from the virtual image to the real image, the first control unit increases the brightness of the first light sources of the first backlight and increases the number of the first light sources to be turned on, which are mounted on the first light-source circuit substrate. Thus, the difference (change) in the brightness when the real image is displayed and when the virtual image is displayed can be further reduced, and the discomfort of the occupant who is a viewer when the display is switched can be suppressed.

[0019] In a fifth aspect depending from the first to fourth aspects, the first control unit may emit the illumination light of the first light source such that the illumination light converges to the first display, and the second control unit may emit the illumination light of the second light source such that the illumination light diverges to the second display.

[0020] In the fifth aspect, the first control unit emits the illumination light of the first light source of the first backlight such that the illumination light converges to the first display, and the second control unit emits the illumination light of the second light source of the second backlight such that the illumination light diverges to the second display. Thus, the light use efficiency can be improved in accordance with the light distribution characteristics required when the real image is displayed and when the virtual image is displayed.

[0021] In a sixth aspect depending from the first to fifth aspects, when the real image is displayed, the first control unit may perform control such that a mounting density of the first light sources to be turned on on an outer edge side in a long-side direction of a light-source circuit mounting substrate formed of a rectangular region on which the first light sources are mounted is made to be larger than a mounting density of the second light sources to be turned on in an inner region of the outer edge side as compared to when the virtual image is displayed by the second control unit.

[0022] In the sixth aspect, when the real image is displayed, the first control unit increases the mounting density of the light sources to be turned on on the outer edge side in the long-side direction of the first light-source circuit substrate on which the first light sources of the first backlight are mounted. Thus, the difference (change) in the brightness when the real image is displayed and when the virtual image is displayed can be further reduced, and the discomfort of the occupant who is a viewer due to the outer edge portion becoming dark, particularly, when the display is switched from the virtual image to the real image can be suppressed.

[0023] A seventh aspect according to the present invention is a control method of a head-up display device that includes: a first backlight including a plurality of first light sources; a first display that causes illumination light of the first light sources to pass therethrough to generate first display light; a second backlight including a plurality of second light sources; a second display that causes illumination light of the second light sources to pass therethrough to generate second display light; an imaging optical system including a first reflection member that reflects the first display light incident from one surface and causes the second display light incident from the other surface to pass therethrough, and a second reflection member that reflects the first display light reflected by the first reflection member or the second display light that has passed through the first reflection member to guide the first display light or the second display light to an emission port, in which the first display light is emitted along a first optical path that is reflected by the first reflection member to generate a real image, and the second display light is emitted along a second optical path that passes through the first reflection member and is reflected by the second reflection member to form a virtual image; a first control unit that controls turning on and off of the plurality of first light sources of the first backlight; and a second control unit that controls turning on and off of the plurality of second light sources of the second backlight, and emits the first display light or the second display light from the emission port toward a translucent member, thereby causing the real image or the virtual image as a display image represented by the first display light or the second display light to be visually recognized, the control method including: when the second control unit displays the virtual image, a step of performing control such that at least some of the second light sources are turned on; and, when the first control unit displays the real image, a step of performing control such that at least some of the first light sources are turned on, and performing control such that power consumption of the first light sources per unit area of the first display is made to be larger than power consumption of the second light sources per unit area of the second display when the virtual image is displayed.

[0024] The seventh aspect is configured to include the first display that causes illumination light emitted from the first light sources of the first backlight for displaying the real image to be reflected by the first reflection member and emits the illumination light toward the translucent member via the second reflection member (first optical path) and generates the real image as the first display light, and the second display that causes illumination light emitted from the second light sources of the second backlight for displaying the virtual image to pass through the first reflection member and emits the illumination light toward the translucent member via the second reflection member (second optical path) and generates the virtual image as the second display light, and to have a control procedure (steps) including, when the second control unit displays the virtual image by switching the imaging optical system to the second optical path, performing control such that at least some of the second light sources are turned on, and, when the first control unit displays the real image by switching the imaging optical system to the first optical path, performing control such that at least some of the first light sources are turned on, and performing control such that the power consumption of the first light sources per unit area of the first display is made to be larger than the power consumption of the second light sources per unit area of the second display when the virtual image is displayed.

[0025] Thus, according to the seventh aspect, when the virtual image is displayed, the second control unit turns on all the second light sources of the second backlight, and, when the real image is displayed, the first control unit turns on all the first light sources of the first backlight and makes the power consumption of the first light sources per unit area larger than that when the virtual image is displayed by increasing the value of the current (current value) flowing through one first light source, so that the brightness when the real image is displayed can be improved, as a result, the difference (change) in the brightness when the real image is displayed and when the virtual image is displayed can be reduced, and the discomfort of the occupant who is a viewer when the display is switched can be suppressed. Moreover, the second control unit may turn on some of the

second light sources of the second backlight when the virtual image is displayed, and the first control unit may turn on all the first light sources of the first backlight when the real image is displayed. In this case, the brightness of the light sources may be the same in the real image display and the virtual image display, or control such that the light sources are turned on more brightly when the real image is displayed may be performed. Furthermore, according to the seventh aspect, lenses in consideration of the light distribution characteristics of both the virtual image and the real image can be optimally designed. Accordingly, appropriate brightness and uniformity can be obtained in both the virtual image display and the real image display, and the display quality can be improved.

[0026] Those skilled in the art will readily understand that the exemplified aspects according to the present invention may be further modified without departing from the spirit of the present invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0027] FIG. 1 is a diagram illustrating an example of a configuration of a head-up display device according to an embodiment of the present invention, which includes an imaging optical system.

[0028] FIGS. 2A and 2B are diagrams illustrating an example of a lens configuration and an optical path between a backlight and a display when a virtual image is displayed in the head-up display device according to the embodiment of the present invention.

[0029] FIGS. 3A and 3B are diagrams illustrating an example of a lens configuration and an optical path between a backlight and a display when a real image is displayed in the head-up display device according to the embodiment of the present invention.

[0030] FIG. 4 is a diagram illustrating an example of a configuration of a control system of the head-up display device according to the embodiment of the present invention.

[0031] FIG. 5 is a flowchart illustrating an example of an operation of the control system of the head-up display device according to the embodiment of the present invention.

[0032] FIGS. 6A and 6B are diagrams illustrating examples of arrangements (layouts) of a plurality of light sources mounted on backlights of the head-up display device according to the embodiment of the present invention.

[0033] FIGS. 7A and 7B are diagrams illustrating examples of light-source lighting patterns of the plurality of light sources mounted on the backlights when the virtual image is displayed and when the real image is displayed of the head-up display device according to the embodiment of the present invention.

[0034] FIGS. 8A and 8B are diagrams illustrating other examples of the light-source lighting patterns of the plurality of light sources mounted on the backlights when the virtual image is displayed and when the real image is displayed of the head-up display device according to the embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

[0035] The best mode described below is used to facilitate understanding of the present invention. Therefore, those skilled in the art should note that the present invention is not unreasonably limited by the embodiment described below (hereinafter, referred to as the present embodiment).

Configuration of Embodiment

[0036] Refer to FIG. 1. FIG. 1 is a diagram illustrating an example of a configuration of a head-up display device (hereinafter, referred to as a HUD device 1 unless otherwise noted) according to the present embodiment, which includes an imaging optical system 2. The HUD device 1 of the present embodiment includes a light source (first light source, refer to 14A in FIG. 6A) that emits white light in a visible wavelength range, is made of, for example, a light-emitting diode (LED), and is

mounted on a light-source circuit substrate (refer to **140A** in FIG. 6A), a backlight **12A** (first backlight) including lenses for condensing light emitted by the light source **14A**, and a display **11A** (first display) such as a liquid crystal display (LCD) panel that causes illumination light of the light source **14A** to pass therethrough to generate first display light **L1** (real image). The light-source circuit substrate **140A** is connected to a control unit **13A** (control circuit substrate), which will be described below, by wiring (not illustrated), and is controlled by the control unit **13A**, so that the LED as the light source **14A** is controlled to be turned on or off.

[0037] Moreover, the HUD device **1** of the present embodiment includes a light source (second light source, refer to **14B** in FIG. 6B) that emits white light in a visible wavelength range and is mounted on a light-source circuit substrate (being a first light-source circuit substrate, refer to **140B** in FIG. 6B), a backlight **12B** (second backlight) including lenses for condensing light emitted by the light source **14B**, and a display **11B** (second display) such as an LCD panel that causes illumination light of the light source **14B** to pass therethrough to generate second display light **L2** (virtual image). The light-source circuit substrate **140B** is connected to a control unit **13B** (control circuit substrate), which will be described below, by wiring (not illustrated), and is controlled by the control unit **13B**, so that the LED as the light source **14B** is controlled to be turned on or off.

[0038] Furthermore, the HUD device **1** of the present embodiment forms the imaging optical system **2** including a first reflection member **22** configured by, for example, a half mirror that reflects the first display light **L1** incident from one surface and causes the second display light **L2** incident from the other surface to pass therethrough, and a second reflection member **24** that reflects the first display light **L1** reflected by the first reflection member **22** or the second display light **L2** that has passed through the first reflection member **22** to guide the first display light **L1** or the second display light **L2** to an emission port **17**, in which the first display light **L1** is emitted along a first optical path **OP1** that is reflected by the first reflection member **22** to generate a real image **V1**, and the second display light **L2** is emitted along a second optical path **OP2** that passes through the first reflection member **22** and is reflected by the second reflection member **24** to form a virtual image.

[0039] Although an infinite number of light beams are originally emitted from the displays **11A** and **11B**, only representative light beams emitted from the centers of the displays **11A** and **11B** and passing through the center of an eye box are indicated by a solid line (real image **V1**) and a broken line (virtual image **V2**) in order to simplify the description. In addition, the first reflection member **22** is not limited to a half mirror, and may be any member as long as it reflects the first display light **L1** on one surface and causes the second display light **L2** to pass therethrough on the other surface. For example, a member to which a wavelength-selective film is attached or a coated transmissive member may be used.

[0040] The first display light **L1** and the second display light **L2** emitted from the displays **11A** and **11B**, respectively, are finally projected toward a windshield **WS**, which is a translucent member, through the emission port **17** that is an opening provided in the upper part of a housing of the HUD device **1**. Accordingly, an occupant **DR** who is a viewer riding a vehicle can visually recognize the real image **V1** formed on the front side of the windshield **WS** (on the vehicle inner side of the windshield **WS**) and the virtual image **V2** formed on the back side of the windshield **WS** (on the vehicle outer side of the windshield **WS**) by visually recognizing the first display light **L1** and the second display light **L2** reflected by the windshield **WS**.

[0041] The HUD device **1** of the present embodiment further has the control units **13A** (first control unit) and **13B** (second control unit) that control switching between the first optical path **OP1** and the second optical path **OP2** of the imaging optical system **2** described above and control turning on and off of a plurality of light sources (not illustrated in FIG. 1) of the backlights **12A** and **12B**. When the virtual image **V2** is displayed by switching the imaging optical system **2** to the second optical path (**OP2**), the control unit **13B** (second control unit) performs control such that at least some of the light sources **14B** are turned on. On the other hand, when the real image **V1** is

displayed by switching the imaging optical system 2 to the first optical path (OP1), the control unit 13A (first control unit) performs control such that at least some of the light sources 14A (first light sources) are turned on, and performs control such that the power consumption of the light sources 14A (first light sources) per unit area of the display 11A (first display) is made to be larger than the power consumption of the light sources 14B (second light sources) per unit area of the display 11B (second display) when the virtual image V2 is displayed.

[0042] Here, “when the real image is displayed, performing control such that the power consumption of the first light sources per unit area of the first display is made to be larger than the power consumption of the second light sources per unit area of the second display when the virtual image is displayed” means, for example, (1) when the real image is displayed, performing control such that the value of the current (current value) flowing through the first light source of the first backlight is increased as compared to when the virtual image is displayed, (2) when the real image is displayed, performing control such that the number of the first light sources to be turned on is increased as compared to when the virtual image is displayed, or (3) when the real image is displayed, performing control such that the current value of the current flowing through the first light source is increased, and performing control such that the number of the first light sources to be turned on is increased as compared to when the virtual image is displayed, or when the real image is displayed, performing control such that the mounting density of the first light sources to be turned on on the outer edge side in the long-side direction of the light-source circuit mounting substrate formed of a rectangular region on which the first light sources are mounted is made to be larger than the mounting density of the second light sources to be turned on in an inner region of the outer edge side as compared to when the virtual image is displayed.

[0043] As will be described below, the control units 13A and 13B can also perform control of display content on the displays 11A and 11B, respectively. Details will be described below.

[0044] Refer to FIGS. 2A, 2B, 3A, and 3B. FIGS. 2A and 2B are diagrams illustrating an example of a lens configuration and the optical path (second optical path OP2) of the display 11B when the virtual image V2 is displayed in the HUD device 1 of the present embodiment, and FIGS. 3A and 3B are diagrams illustrating an example of a lens configuration and the optical path (first optical path OP1) of the display 11A when the real image V1 is displayed in the HUD device 1 of the present embodiment.

[0045] An illumination optical system for displaying the real image V1 and an illumination optical system for displaying the virtual image V2 used in the HUD device 1 of the present embodiment are independently controlled by the control units 13A and 13B, respectively, and thus optimized and designed exclusively for the control units 13A and 13B. Lenses of the illumination optical system for displaying the real image V1 and the illumination optical system for displaying the virtual image V2 are designed so as to satisfy both the light distribution characteristics in which the optical axis narrows in both the vertical direction (V) and the horizontal direction (H) required when the real image V1 is displayed (the light source of the backlight 12A emits illumination light such that the illumination light converges to the display) and the light distribution characteristics in which the optical axis spreads in both the vertical direction (V) and the horizontal direction (H) required when the virtual image V2 is displayed (the light source of the backlight 12B emits illumination light such that the illumination light diverges to the display) (for example, a condenser lens 123 and lenticular lenses 124 and 125 illustrated in FIGS. 2A and 2B, which are included in an illumination optical system and will be described below), and excellent brightness and uniformity can be obtained in both a visual recognition state of the virtual image V2 and a visual recognition state of the real image V1.

[0046] FIG. 2A illustrates the lens configuration and the optical path in the H direction, and FIG. 2B illustrates the lens configuration and the optical path in the V direction when the virtual image V2 is displayed. In FIGS. 2A and 2B, light emitted from an LED (122 in this case) serving as the light source 14B is converted into parallel light which is substantially parallel to the optical axis in

both the H direction and the V direction by the condenser lens **123**. A light receiving surface of the first lenticular lens **124** on the side of the light source **122** has a structure in which a plurality of cylindrical lenses having a convex curved surface on the side of the LED **122** in a cross section in the V direction are arranged in parallel along the V direction. An exit surface of the first lenticular lens **124** has a structure in which a plurality of cylindrical lenses having a convex curved surface on the exit side of the second display light L2 in a cross section in the V direction are arranged in parallel along the V direction. With such a structure, the first lenticular lens **124** condenses the light of the LED **122** in the V direction to form a multiple image.

[0047] A light receiving surface of the second lenticular lens **125** on the side of the LED **122** has a structure in which a plurality of cylindrical lenses having a convex curved surface on the side of the LED **122** in a cross section in the H direction are arranged along the H direction. An exit surface of the second lenticular lens **125** is a toroidal surface having a concave shape in both the V direction and the H direction. With such a configuration, the second lenticular lens **125** condenses the light of the LED **122** in the H direction to form a multiple image, and aligns the direction of the light with the subsequent imaging optical system **2**. The second lenticular lens **125** for displaying the virtual image V2 is referred to as a first lens **125a**.

[0048] The second display light L2 emitted from the first lens **125a** is diffused by a diffuser plate **127** (not illustrated in FIG. 1) to reduce brightness unevenness and the like. The display light L2 diffused by the diffuser plate **127** is incident on an LCD panel **126**, which is the display **11B** (refer to FIG. 1), to generate an image in accordance with the control of a control board **162** (the control unit **13B** in FIG. 1), and is emitted to the subsequent first reflection member **22**. With such a lens configuration and an optical path, the occupant DR can visually recognize the virtual image V2 with appropriate brightness and uniformity.

[0049] On the other hand, FIGS. 3A and 3B are diagrams illustrating an example of a lens configuration and the optical path (OP1) when the real image V1 is displayed in the HUD device **1** of the present embodiment. FIG. 3A illustrates the lens configuration and the optical path in the H direction, and FIG. 3B illustrates the lens configuration and the optical path in the V direction. In FIGS. 3A and 3B, the condenser lens **123** and the first lenticular lens **124** have the same configurations as those in FIGS. 2A and 2B. That is, the condenser lens **123** converts the light emitted from the LED **122** into parallel light in both the H direction and the V direction, and the first lenticular lens **124** condenses the light of the LED **122** in the V direction to form a multiple image.

[0050] Also, regarding the second lenticular lens **125**, the function of condensing the light of the LED **122** in the H direction to form a multiple image is the same as that in the case of FIGS. 2A and 2B. Therefore, as in the case of FIGS. 2A and 2B, the light receiving surface on the side of the LED **122** has a structure in which a plurality of cylindrical lenses having a convex curved surface on the side of the LED **122** in a cross section in the H direction are arranged along the H direction. On the other hand, in order to realize the light distribution characteristics (the light source **14A** for displaying a real image (the LED **122** in this case) emits illumination light such that the illumination light converges to the display **11A** for displaying the real image V1, and the light source **14B** for displaying a virtual image emits illumination light such that the illumination light diverges to the display **11B** for displaying the virtual image V2), the exit surface is a toroidal surface having a convex shape in both the V direction and the H direction. By forming the exit surface into a convex shape, the light distribution characteristics are narrowed in both the H direction and the V direction. The second lenticular lens **125** for displaying the real image V1 is referred to as a second lens **125b**.

[0051] The first display light L1 emitted from the second lens **125b** is diffused by the diffuser plate **127** (not illustrated in FIG. 1) to reduce brightness unevenness and the like, is incident on the display **11A** (the LCD panel **126** in this case) to generate an image (first display light L1) in accordance with the control of the control board **162**, and is emitted to the subsequent first

reflection member **22**. As illustrated in FIGS. **3A** and **3B**, when the real image **V1** is displayed, in order to generate the real image **V1**, the optical axes in the H direction and the V direction are made to intersect between the windshield **WS** and the viewpoint **EB** (eye box) of the occupant **DR** who is a viewer. In this case, the intersection of the optical axes can be freely set in accordance with the position and magnification of the real image **V1** to be displayed. With such a lens configuration and an optical path, the occupant **DR** can visually recognize the real image **V1** with appropriate brightness and uniformity.

[0052] Refer to FIG. **4**. FIG. **4** is a diagram illustrating an example of a configuration of a control system of the HUD device **1** of the present embodiment, specifically, is a functional configuration diagram illustrating a configuration of a picture generation unit **PGU-1** (**PGU 10A**), and includes the control unit **13A** illustrated in FIG. **1**. The HUD device **1** of the present embodiment further has a **PGU-2** (**PGU 10B**), and also includes the control unit **13B** illustrated in FIG. **1**. The picture generation unit refers to a control system for image generation and display, only a minimum necessary configuration directly related to the HUD device **1** of the present embodiment is illustrated in FIG. **4**, and other known configurations are omitted.

[0053] In FIG. **4**, the **PGU-1** (**PGU 10A**) includes the display **11A**, the backlight **12A** (light source **14A**), and the control unit **13A**. The control unit **13A** includes a display control unit **131** that issues a command to the display **11A** to generate the first display light **L1** (real image **V1**) representing a display image on the basis of information or a signal transmitted from various devices **30** such as a vehicle speed sensor, a navigation device, a radio detecting and ranging (**RADAR**), and a light detection and ranging (**LiDAR**).

[0054] Moreover, the control unit **13A** includes a display drive unit **132** that generates an image with the light emitted from the light source **14A** of the backlight **12A**, switches the polarization of light to be emitted to a first polarized light state and a second polarized light state different from each other, and performs switching control of the polarizing direction of the display **11A** that generates the first display light **L1** representing a display image, on the basis of a signal transmitted from a switch **20** for switching, for example, a vehicle driving mode (manual driving, automatic driving).

[0055] Furthermore, the control unit **13A** includes a light source drive unit **133** that controls power to be supplied necessary for turning on and off the light source **14A** of the backlight **12A**, which is mounted on the light-source circuit substrate **140A** (refer to FIG. **6A**). The light source drive unit **133** controls ON/OFF timing of each LED as the light source **14A**, which is mounted on the light-source circuit substrate **140A**, and the value of the current (current value) flowing through each LED, and also controls the voltage supplied to the light source **14A**, so that the power efficiency can be improved. When the real image **VI** is displayed by switching the imaging optical system **2** to the **OP1** (first optical path), the control unit **13A** can perform control such that at least some of the light sources **14A** (first light sources) of the backlight **12A** are turned on, and can perform control such that the power consumption of the light sources **14A** (first light sources) per unit area of the display **11A** (first display) is made to be larger than the power consumption of the light sources **14B** (second light sources) per unit area of the display **11B** (second display) when the virtual image **V2** is displayed.

[0056] A display unit **11A** (same as the display **11A**) includes a thin film transistor (**TFT**) type display element **111** that forms the first display light **L1** representing a figure having a freely-selected shape on the basis of a signal transmitted from the display control unit **131**. For example, during automatic driving, the display drive unit **132** performs switching control. At this time, the display control unit **131** controls the display element **111** to generate the display light **L1** representing an assistant or an agent that supports the driving of the occupant **DR**, a character indicating the assistant or the agent, or the like.

[0057] The **PGU-2** (**10B**) also has the same configuration as that of the **PGU-1** (**10A**), and includes a display unit **11B** (same as the display **11B**), the backlight **12B** (light source **14B**), and the control

unit **13B** (all are not illustrated). The control unit **13B** has the display control unit **131**, the display drive unit **132**, and the light source drive unit **133**. The display control unit **131** controls the display element **111** to generate the display light **L2** representing vehicle information, route guidance information, warning display, and the like. The light source drive unit **133** performs lighting control of the second light source **14B** of the second backlight **12B**, and can perform control, when the virtual image **V2** is displayed, such that at least some of the second light sources **14B** of the second backlight **12B** are turned on.

[0058] As described above, in the HUD device **1** of the present embodiment, when the real image **V1** is to be displayed, the control unit **13A** switches the imaging optical system **2** to the optical path **OP1** (first optical path) in which the first display light **L1** (real image **V1**) generated by the display unit **11A** (same as the display **11A** in FIG. **1**) is reflected by the first reflection member **22**, so that the first display light **L1** is further reflected by the second reflection member **24** to be incident on the windshield **WS**, which is a translucent member, and is reflected by the windshield **WS** to form the real image **V1** that is displayable on the vehicle inner side of the windshield **WS** in front of the occupant **DR** (in an imaging area virtually set in a state of standing vertically with respect to the road surface). On the other hand, in the HUD device **1** of the present embodiment, when the virtual image **V2** is to be displayed, the control unit **13B** switches the imaging optical system **2** to the optical path **OP2** (second optical path) in which the display light **L2** generated by the display **11B** is caused to pass through the first reflection member **22**, so that the display light **L2** is further reflected by the second reflection member **24** to be incident on the windshield **WS**, which is a translucent member, and is reflected by the windshield **WS** to form the virtual image **V2** that is displayable on the vehicle outer side of the windshield **WS** (in an imaging area virtually set in front of the vehicle and set in a state of being inclined with respect to the road surface).

[0059] That is, for example, the imaging area of the real image **V1** having an angle of 45 degrees or more with respect to the road surface is assumed to be used in a scene such as viewing entertainment content during automatic driving or stopping, and there is an advantage that the visibility is improved due to the real image **V1** being displayed in a state of standing with respect to the road surface. On the other hand, for example, in the imaging area of the virtual image **V2** having an angle of less than 45 degrees with respect to the road surface, the display content appears to be developed on the road surface. Therefore, when navigation or the like is performed, the display content appears to be superimposed on the road surface, and there is an advantage that intuitive information presentation is possible.

Operation of Embodiment

[0060] Refer to FIG. **5**. FIG. **5** is a flowchart illustrating an example of an operation of the control system of the HUD device **1** of the present embodiment. In addition, FIGS. **6A** and **6B** are diagrams illustrating examples of arrangements (layouts) of the plurality of light sources **14A** and **14B** mounted on the backlights **12A** and **12B** of the HUD device **1** of the present embodiment. Hereinafter, with reference to FIGS. **5**, **6A**, and **6B**, the operation of the control system (the PGU **10A** and the PGU **10B**, mainly the control unit **13A** and the control unit **13B**) of the HUD device **1** of the present embodiment illustrated in FIG. **4** will be described in detail.

[0061] In the PGU **10A**, first, the control unit **13B** (display drive unit **132**) determines whether the vehicle is in manual driving or in automatic driving on the basis of a signal transmitted from the switch **20** for switching a vehicle driving mode (manual driving, automatic driving) (step **ST101**). When it is determined that the vehicle is in manual driving (step **ST101** “M”), the control unit **13B** (display control unit **131**) controls the display element **111** of the display unit **11B** (display **11B**) to generate the display light **L2** (virtual image **V2**) representing vehicle information, route guidance information, warning display, and the like (step **ST102**).

[0062] Subsequently, for example, as illustrated in FIG. **6B** as an example of the arrangement (layout) of the light sources **14B** on the light-source circuit substrate **140B**, the control unit **13B** (light source drive unit **133**) performs control such that at least some of the LEDs (all the LEDs in

this case) as the light sources **14B** are turned on (step **ST103**), and the display unit **11B** emits (projects) the generated first display light **L1** (real image) toward the windshield **WS**, which is a translucent member, via the imaging optical system **2** (optical path **OP1**) and the emission port **17** (step **ST106**).

[0063] On the other hand, in step **ST101**, when the vehicle is in automatic driving (step **ST101** “A”), the control unit **13A** (display drive unit **132**) drive-controls the display element **111** to generate the display light **L1** representing an assistant or an agent that supports the driving of the occupant **DR**, a character indicating the assistant or the agent, or the like (step **ST104**).

[0064] Next, for example, as illustrated in FIG. **6A** as an example of the arrangement (layout) of the light sources **14A** on the light-source circuit substrate **140A**, the control unit **13A** (light source drive unit **133**) performs control such that at least some of the LEDs (all the LEDs in this case) as the light sources **14A** are turned on, and also performs control such that power to be supplied to the first light sources **14A** mounted on the light-source circuit substrate **140A** of the first backlight **12A** is increased as compared to when the virtual image is displayed (step **ST105**). When the power to be supplied to the light sources **14A** are controlled, the power consumption is increased, and the brightness when the real image **V1** is displayed is made to be larger than that when the virtual image **V2** is displayed, for example, by performing control such that the value of the current (current value) flowing through the first light sources **14A** of the first backlight **12A** is increased, by performing control such that the number of the first light sources **14A** to be turned on (lighting area) is increased, or by performing both.

[0065] Moreover, for example, control such that only some of the second light sources **14B** of the second backlight **12B** are turned on (indicated as ON) when the virtual image **V2** is displayed, as illustrated in FIG. **7B**, and such that all the first light sources **14A** of the first backlight **12A** are turned on when the real image **V1** is displayed, as illustrated in FIG. **7A**, may be performed. In this case, the brightness of the light sources (**14A**, **14B**) may be the same when the virtual image **V2** is displayed and when the real image **V1** is displayed, or the light sources may be turned on more brightly when the real image **V1** is displayed. Furthermore, when the real image **V1** is displayed, as illustrated in FIG. **8A**, the number of the first light sources **14A** to be turned on, which are mounted on the outer edge portion of the light-source circuit substrate **140A**, may be increased (indicated as “dense” in the drawing) as compared to when the virtual image is displayed, as illustrated in FIG. **8B**. Lighting patterns of the light sources illustrated in FIGS. **7A**, **7B**, **8A**, and **8B** will be described below in detail.

[0066] Finally, the display unit **11A** emits (projects) the generated first display light **L1** (real image **V1**) toward the windshield **WS**, which is a translucent member, via the imaging optical system **2** (optical path **OP1**) and the emission port **17** (step **ST106**).

[0067] FIGS. **7A** and **7B** illustrate examples of the lighting patterns of the light sources **14A** and the light sources **14B** mounted on the backlight **12A** and the backlight **12B** when the real image **V1** is displayed (FIG. **7A**) and when the virtual image **V2** is displayed (FIG. **7B**) of the HUD device **1** of the present embodiment. In FIG. **7B**, a light source mounted region surrounded by a broken line and indicated as ON is turned on, and a light source mounted region surrounded by a broken line and indicated as OFF is turned off.

[0068] In the lighting pattern of the light sources **14B** of the backlight **12B** when the virtual image **V2** is displayed as illustrated in FIG. **7B**, the light sources **14B** laid out in two rows on each of the left and right sides of the outer edge portion in the long-side direction (horizontal direction) of the rectangular light-source circuit substrate **140B** are turned off (indicated as OFF), and the light sources **14B** laid out therebetween are turned on (indicated as ON). On the other hand, all the first light sources **14A** of the backlight **12A** when the real image **V1** is displayed as illustrated in FIG. **7A** are turned on. The brightness of each light source **14B** of the backlight **12B** in the lighting pattern of the second light sources **14B** when the virtual image **V2** is displayed as illustrated in FIG. **7B** may be the same as that when the real image **V1** is displayed as illustrated in FIG. **7A**.

[0069] FIGS. 8A and 8B illustrate other examples of the lighting patterns of the light sources **14A** and the light sources **14B** mounted on the backlight **12A** and the backlight **12B** when the real image **V1** is displayed (FIG. 8A) and when the virtual image **V2** is displayed (FIG. 8B) of the HUD device **1** of the present embodiment. In the light-source lighting pattern when the real image **V1** is displayed as illustrated in FIG. 8A, the light sources **14A** arranged and turned on in three rows on each of the left and right sides of the edge portion in the long-side direction (horizontal direction) of the rectangular light-source circuit substrate **140A** are “densely” mounted, and the light sources **14A** arranged therebetween are “sparsely” mounted (indicated as “coarse”). On the other hand, the light sources **14B** when the virtual image **V2** is displayed as illustrated in FIG. 8B are “sparsely” mounted on the light-source circuit substrate **140B** (indicated as “coarse”) as a whole.

[0070] As described above, by devising the number of the light sources **14A** and **14B** to be turned on (FIGS. 7A and 7B) and the arrangement of the lighted light sources **14A** and **14B** (FIGS. 8A and 8B), a difference in brightness (brightness difference) of the display image between when the real image **V1** is displayed and when the virtual image **V2** is displayed can be reduced, and the discomfort of the occupant **DR** due to a change in brightness (brightness change) of the display images during display switching between the virtual image **V2** and the real image **V1** can be suppressed.

Effect of Invention

[0071] As described above, for example, as illustrated in FIG. 1, a head-up display device of the present embodiment is the HUD device **1** that has the emission port **17** and emits the first display light **L1** and the second display light **L2** from the emission port **17** toward the translucent member **WS**, thereby causing the real image **V1** and the virtual image **V2** as display images represented by the first display light **L1** and the second display light **L2** to be visually recognized. The HUD device **1** includes the first backlight **12A** including the first light source (refer to **14A** in FIG. 6A), the first display **11A** that causes illumination light of the first light source **14A** to pass therethrough to generate the first display light **L1**, the second backlight **12B** including the second light source (refer to **14B** in FIG. 6B), the second display **11B** that causes illumination light of the second light source **14B** to pass therethrough to generate the second display light **L2**, the imaging optical system **2** including the first reflection member **22** that reflects the first display light **L1** incident from one surface and causes the second display light **L2** incident from the other surface to pass therethrough, and the second reflection member **24** that reflects the first display light **L1** reflected by the first reflection member **22** or the second display light **L2** that has passed through the first reflection member **22** to guide the first display light **L1** or the second display light **L2** to the emission port **17**, in which the first display light **L1** is emitted along the first optical path **OP1** that is reflected by the first reflection member **22** to generate the real image **V1**, and the second display light **L2** is emitted along the second optical path **OP2** that passes through the first reflection member **22** and is reflected by the second reflection member **24** to form the virtual image **V2**, the first control unit **13A** that controls turning on and off of a plurality of the first light sources **14A** of the first backlight **12A**, and the second control unit **13B** that controls turning on and off of a plurality of the second light sources **14B** of the second backlight **12B**, in which, when the virtual image **V2** is displayed, the second control unit **13B** performs control such that at least some of the second light sources **14B** are turned on, and, when the real image **VI** is displayed, the first control unit **13A** performs control such that at least some of the first light sources **14A** are turned on, and performs control such that the power consumption of the first light sources **14A** per unit area of the first display **11A** is made to be larger than the power consumption of the second light sources **14B** per unit area of the second display **11B** when the virtual image **V2** is displayed.

[0072] The HUD device **1** of the present embodiment is configured to include the first display **11A** that causes illumination light emitted from the first light source **14A** of the first backlight **12A** for displaying the real image **V1** to be reflected by the first reflection member **22** and emits the illumination light toward the translucent member **WS** via the second reflection member **24** (first

optical path OP1) and generates the real image V1 as the first display light L1, and the second display 11B that causes illumination light emitted from the second light source 14B of the second backlight 12B for displaying the virtual image V2 to pass through the first reflection member 22 and emits the illumination light toward the translucent member WS via the second reflection member 24 (second optical path OP2) and generates the virtual image V2 as the second display light L2, in which, when the virtual image V2 is displayed, the second control unit 13B performs control such that at least some of the second light sources 14B are turned on, and, when the real image V1 is displayed, the first control unit 13A performs control such that at least some of the first light sources 14A are turned on, and performs control such that the power consumption of the first light sources 14A per unit area of the first display 11A is made to be larger than the power consumption of the second light sources 14B per unit area of the second display 11B when the virtual image V2 is displayed.

[0073] According to the HUD device 1 of the present embodiment, when the virtual image V2 is displayed, the second control unit 13B turns on all the second light sources 14B of the second backlight 12B, and, when the real image V1 is displayed, the first control unit 13A turns on all the first light sources 14A of the first backlight 12A and performs control such that the power consumption of the first light sources 14A per unit area is made to be larger than that when the virtual image V2 is displayed by, for example, increasing the value of the current (current value) flowing through one first light source 14A, so that the brightness of the display image when the real image V1 is displayed can be increased, as a result, the difference (change) in the brightness of the display image when the display is switched to the virtual image V2 or the real image V1 can be reduced, and the discomfort of the occupant DR who is a viewer when the display is switched can be suppressed.

[0074] Moreover, the second control unit 13B may turn on some of the second light sources 14B of the second backlight 12B when the virtual image V2 is displayed, and the first control unit 13A may turn on all the first light sources 14A of the first backlight 12A when the real image V1 is displayed. In this case, the brightness of the light sources may be the same in the real image V1 display and the virtual image V2 display, or control such that the light sources are turned on more brightly when the real image V1 is displayed may be performed. Furthermore, according to the HUD device 1 of the present embodiment, lenses in consideration of the light distribution characteristics of both the virtual image V2 and the real image V1 can be optimally designed. Accordingly, appropriate brightness and uniformity can be obtained in both the virtual image V2 display and the real image V1 display, and the display quality can be improved.

[0075] According to the HUD device 1 of the present embodiment, when the real image V1 is displayed, the first control unit 13A performs control such that the value of the current (current value) flowing through the first light source 14A of the first backlight 12A is increased as compared to when the virtual image V2 is displayed by the second control unit 13B. Thus, particularly, when the display is switched from the virtual image V2 to the real image V1, the difference (change) in the brightness of the display image from when the virtual image V2 is displayed can be reduced without changing the lighting area of the first light sources 14A mounted on the first light-source circuit substrate 140A of the first backlight 12A, and the discomfort of the occupant DR who is a viewer when the display is switched can be suppressed.

[0076] According to the HUD device 1 of the present embodiment, when the display is switched from the virtual image V2 to the real image V1, the first control unit 13A increases the number of the first light sources 14A to be turned on, which are mounted on the first light-source circuit substrate 140A of the first backlight 12A. Thus, the difference (change) in the brightness of the display images when the real image V1 is displayed and when the virtual image V2 is displayed can be reduced, and the discomfort of the occupant DR who is a viewer when the display is switched can be suppressed.

[0077] According to the HUD device 1 of the present embodiment, when the display is switched

from the virtual image V2 to the real image V1, the first control unit 13A performs control such that the brightness of the first light sources 14A of the first backlight 12A is increased and the number of the first light sources 14A to be turned on, which are mounted on the first light-source circuit substrate 140A, is increased. Thus, the difference (change) in the brightness of the display images when the real image V1 is displayed and when the virtual image V2 is displayed can be further reduced, and the discomfort of the occupant DR who is a viewer when the display is switched can be suppressed.

[0078] According to the HUD device 1 of the present embodiment, the first control unit 13A emits the illumination light of the first light source 14A of the first backlight 12A such that the illumination light converges to the first display 11A, and the second control unit 13B emits the illumination light of the second light source 14B of the second backlight 12B such that the illumination light diverges to the second display 11B. Thus, the light use efficiency can be improved in accordance with the light distribution characteristics required when the real image V1 is displayed and when the virtual image V2 is displayed.

[0079] According to the HUD device 1 of the present embodiment, when the real image V1 is displayed, the first control unit 13A increases the mounting density of the light sources 14A to be turned on on the outer edge side in the long-side direction of the first light-source circuit substrate 140A on which the first light sources 14A of the first backlight 12A are mounted.

[0080] Thus, the difference (change) in the brightness of the display images when the real image V1 is displayed and when the virtual image V2 is displayed can be further reduced, and the discomfort of the occupant DR who is a viewer due to the outer edge portion becoming dark, particularly, when the display is switched from the virtual image V2 to the real image V1 can be suppressed.

[0081] A control method of a head-up display device of the present embodiment is a control method of the HUD device 1 that includes, for example, as illustrated in FIG. 1, the first backlight 12A including a plurality of first light sources 14A, the first display 11A that causes illumination light of the first light sources 14A to pass therethrough to generate the first display light L1, the second backlight 12B including a plurality of second light sources 14B, the second display 11B that causes illumination light of the second light sources 14B to pass therethrough to generate the second display light L2, the imaging optical system 2 including the first reflection member 22 that reflects the first display light L1 incident from one surface and causes the second display light L2 incident from the other surface to pass therethrough, and the second reflection member 24 that reflects the first display light L1 reflected by the first reflection member 22 or the second display light L2 that has passed through the first reflection member 22 to guide the first display light L1 or the second display light L2 to the emission port 17, in which the first display light L1 is emitted along the first optical path OP1 that is reflected by the first reflection member 22 to generate the real image V1, and the second display light L2 is emitted along the second optical path OP2 that passes through the first reflection member 22 and is reflected by the second reflection member 24 to form the virtual image V2, the first control unit 13A that controls turning on and off of the plurality of first light sources 14A of the first backlight 12A, and the second control unit 13B that controls turning on and off of the plurality of second light sources 14B of the second backlight 12B, and emits the first display light L1 or the second display light L2 from the emission port 17 toward the translucent member WS, thereby causing the real image V1 or the virtual image V2 as a display image represented by the first display light L1 or the second display light L2 to be visually recognized. For example, as illustrated in FIG. 5, the control method includes, when the second control unit 13B displays the virtual image V2, a step of performing control such that at least some of the second light sources 14B are turned on (refer to ST101 “M” to ST103), and, when the first control unit 13A displays the real image V1, a step of performing control such that at least some of the first light sources 14A are turned on, and performing control such that the power consumption of the first light sources 14A per unit area of the first display 11A is made to be larger than the

power consumption of the second light sources **14B** per unit area of the second display **11B** when the virtual image **V2** is displayed (refer to **ST101** “A,” **ST104**, and **ST105**).

[0082] The control method of the HUD device **1** of the present embodiment is configured to include the first display **11A** that causes illumination light emitted from the first light sources **14A** of the first backlight **12A** for displaying the real image **V1** to be reflected by the first reflection member **22** and emits the illumination light toward the translucent member **WS** via the second reflection member **22** (first optical path **OP1**) and generates the real image **V1** as the first display light **L1**, and the second display **11B** that causes illumination light emitted from the second light sources **14B** of the second backlight **12B** for displaying the virtual image **V2** to pass through the first reflection member **22** and emits the illumination light toward the translucent member **WS** via the second reflection member **24** (second optical path **OP2**) and generates the virtual image **V2** as the second display light **L2**, and to have a control procedure (refer to **FIG. 5**) including, when the second control unit **13B** displays the virtual image **V2**, performing control such that at least some of the second light sources **14B** are turned on, and, when the first control unit **13A** displays the real image **V1**, performing control such that at least some of the first light sources **14A** are turned on, and performing control such that the power consumption of the first light sources **14A** per unit area of the first display **11A** is made to be larger than the power consumption of the second light sources **14B** per unit area of the second display **11B** when the virtual image **V2** is displayed.

[0083] Thus, according to the control method of the HUD device **1** of the present embodiment, when the virtual image **V2** is displayed, the second control unit **13B** turns on all the second light sources **14B** of the second backlight **12B**, and, when the real image **V1** is displayed, the first control unit **13A** turns on all the first light sources **14A** of the first backlight **12A** and makes the power consumption of the first light sources **14A** per unit area larger than that when the virtual image **V2** is displayed by increasing the value of the current (current value) flowing through one first light source **14A**, so that the brightness when the real image **V1** is displayed can be improved, as a result, the difference (change) in the brightness of the display images when the real image **V1** is displayed and when the virtual image **V2** is displayed can be reduced, and the discomfort of the occupant **DR** who is a viewer when the display is switched can be suppressed. Moreover, the second control unit **13B** may turn on some of the second light sources **14B** of the second backlight **12B** when the virtual image **V2** is displayed, and the first control unit **13A** may turn on all the first light sources **14A** of the first backlight **12A** when the real image **V1** is displayed. In this case, the brightness of the light sources may be the same in the real image **V1** display and the virtual image **V2** display, or control such that the light sources are turned on more brightly when the real image **V1** is displayed may be performed. Furthermore, according to the control method of the HUD device **1** of the present embodiment, lenses in consideration of the light distribution characteristics of both the virtual image **V2** and the real image **V1** can be optimally designed. Accordingly, appropriate brightness and uniformity can be obtained in both the virtual image **V2** display and the real image **V1** display, and the display quality can be improved.

[0084] Although the windshield **WS** is used as a translucent member in the present embodiment, a flat glass or a combiner may be used.

[0085] The present invention is not limited to the above-described exemplary embodiment, and those skilled in the art can easily modify the above-described exemplary embodiment to the extent that the above-described exemplary embodiment is included in the scope of the claims.

REFERENCE SIGNS LIST

[0086] **1** head-up display device (HUD device) [0087] **2** imaging optical system [0088] **10A** picture generation unit (PGU-1) [0089] **10B** picture generation unit (PGU-2) [0090] **11A** first display [0091] **11B** second display [0092] **12A** first backlight [0093] **12B** second backlight [0094] **13A** first control unit [0095] **13B** second control unit [0096] **14A** first light source [0097] **14B** second light source [0098] **17** emission port [0099] **20** switch [0100] **30** various devices [0101] **22** first reflection member [0102] **24** second reflection member [0103] **111** display element [0104] **131**

display control unit [0105] **132** display drive unit [0106] **133** light source drive unit [0107] **140A** first light-source circuit substrate [0108] **140B** second light-source circuit substrate [0109] **V1** real image [0110] **V2** virtual image [0111] **L1** first display light [0112] **L2** second display light [0113] **OP1** first optical path [0114] **OP2** second optical path

Claims

1. A head-up display device that has an emission port and emits display light from the emission port toward a translucent member, thereby causing a real image and a virtual image as display images represented by the display light to be visually recognized, the head-up display device comprising: a first backlight including a first light source; a first display that causes illumination light of the first light source to pass therethrough to generate first display light; a second backlight including a second light source; a second display that causes illumination light of the second light source to pass therethrough to generate second display light; an imaging optical system including a first reflection member that reflects the first display light incident from one surface and causes the second display light incident from the other surface to pass therethrough, and a second reflection member that reflects the first display light reflected by the first reflection member or the second display light that has passed through the first reflection member to guide the first display light or the second display light to the emission port, wherein the first display light is emitted along a first optical path that is reflected by the first reflection member to generate the real image, and the second display light is emitted along a second optical path that passes through the first reflection member and is reflected by the second reflection member to form the virtual image; a first control unit that controls turning on and off of a plurality of the first light sources of the first backlight; and a second control unit that controls turning on and off of a plurality of the second light sources of the second backlight, wherein, when displaying the virtual image, the second control unit performs control such that at least some of the second light sources are turned on, and, when displaying the real image, the first control unit performs control such that at least some of the first light sources are turned on, and performs control such that power consumption of the first light sources per unit area of the first display is made to be larger than power consumption of the second light sources per unit area of the second display when the virtual image is displayed.
2. The head-up display device according to claim 1, wherein, when displaying the real image, the first control unit performs control such that a current value of a current flowing through the first light source is increased as compared to when the virtual image is displayed by the second control unit.
3. The head-up display device according to claim 1, wherein, when displaying the real image, the first control unit performs control such that the number of the first light sources to be turned on is increased as compared to when the virtual image is displayed by the second control unit.
4. The head-up display device according to claim 1, wherein, when displaying the real image, the first control unit performs control such that a current value of a current flowing through the first light source is increased, and performs control such that the number of the first light sources to be turned on is increased as compared to when the virtual image is displayed by the second control unit.
5. The head-up display device according to claim 1, wherein the first control unit emits the illumination light of the first light source such that the illumination light converges to the first display, and the second control unit emits the illumination light of the second light source such that the illumination light diverges to the second display.
6. The head-up display device according to claim 1, wherein, when displaying the real image, the first control unit performs control such that a mounting density of the first light sources to be turned on on an outer edge side in a long-side direction of a light-source circuit mounting substrate formed of a rectangular region on which the first light sources are mounted is made to be larger

than a mounting density of the second light sources to be turned on in an inner region of the outer edge side as compared to when the virtual image is displayed by the second control unit.

7. A control method of a head-up display device that includes: a first backlight including a plurality of first light sources; a first display that causes illumination light of the first light sources to pass therethrough to generate first display light; a second backlight including a plurality of second light sources; a second display that causes illumination light of the second light sources to pass therethrough to generate second display light; an imaging optical system including a first reflection member that reflects the first display light incident from one surface and causes the second display light incident from the other surface to pass therethrough, and a second reflection member that reflects the first display light reflected by the first reflection member or the second display light that has passed through the first reflection member to guide the first display light or the second display light to an emission port, wherein the first display light is emitted along a first optical path that is reflected by the first reflection member to generate a real image, and the second display light is emitted along a second optical path that passes through the first reflection member and is reflected by the second reflection member to form a virtual image; a first control unit that controls turning on and off of the plurality of first light sources of the first backlight; and a second control unit that controls turning on and off of the plurality of second light sources of the second backlight, and emits the first display light or the second display light from the emission port toward a translucent member, thereby causing the real image or the virtual image as a display image represented by the first display light or the second display light to be visually recognized, the control method comprising: when the second control unit displays the virtual image, a step of performing control such that at least some of the second light sources are turned on; and, when the first control unit displays the real image, a step of performing control such that at least some of the first light sources are turned on, and performing control such that power consumption of the first light sources per unit area of the first display is made to be larger than power consumption of the second light sources per unit area of the second display when the virtual image is displayed.
